MAT 017B Final Exam

September 13, 2018

Name: Key

SID:

$$\int x^n\,dx = \frac{x^{n+1}}{n+1} + C,\ n \neq -1 \qquad \int \frac{1}{x}\,dx = \ln|x| + C \qquad \int e^x\,dx = e^x + C$$

$$\int \sin x\,dx = -\cos x + C \qquad \int \cos x\,dx = \sin x + C \qquad \int \sec^2 x\,dx = \tan x + C$$

$$\int \frac{1}{x^2+a^2}\,dx = \frac{1}{a}\tan^{-1}\left(\frac{x}{a}\right) + C \qquad \int_a^b f(x)\,dx = F(b) - F(a)$$

$$\int_a^b f(x)\,dx \approx \sum_{k=1}^{n} f\left(x_k\right)\Delta x, \quad \Delta x = \frac{b-a}{n}, \quad x_k = a + k\Delta x$$

$$\sum_{k=1}^{n} 1 = n \qquad \sum_{k=1}^{n} k = \frac{n\left(n+1\right)}{2} \qquad \sum_{k=1}^{n} k^2 = \frac{n\left(n+1\right)\left(2n+1\right)}{6}$$

$$\frac{d}{dx}\int_a^x f(t)\,dt = f(x) \qquad \frac{d}{dx}\int_{h(x)}^{g(x)} f(t)\,dt = f[g(x)]g'(x) - f[h(x)]h'(x)$$

$$\int f'(g(x))g'(x)\,dx = f(g(x)) + C \qquad \int u\,dv = uv - \int v\,du$$

$$A = \int [f(x) - g(x)]\,dx \quad \Delta y = \int \frac{dy}{dx}\,dx \quad \bar{y} = \frac{1}{b-a}\int_a^b f(x)\,dx$$

$$f\left(x\right) \approx \sum_{k=0}^{n} \frac{f^{(k)}\left(a\right)}{k!}\left(x-a\right)^k \quad \frac{dy}{dx} = f(x)g(y) \quad \int \frac{dy}{g(y)} = \int f(x)\,dx$$

$$y' + p(x)y = q(x) \quad I\left(x\right) = e^{\int p(x)dx} \quad y = \frac{1}{I(x)}\int I(x)q(x)\,dx$$

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \quad \Delta = \det\left(A\right) = ad - bc \quad \tau = \operatorname{tr}\left(A\right) = a + d$$

$$A\vec{x} = \lambda\vec{x} \quad \det\left(A - \lambda I\right) = 0 \quad \lambda^2 - \tau\lambda + \Delta = 0 \quad \left(A - \lambda I\right)\vec{x} = 0$$

$$|\vec{x}| = \sqrt{x_1^2 + x_2^2 + \cdots + x_n^2} \quad \vec{x}\cdot\vec{y} = \sum_{k=1}^{n} x_k y_k = |\vec{x}||\vec{y}|\cos\theta \quad \hat{x} = \frac{\vec{x}}{|\vec{x}|}$$

1. (15 points) Let x and y be the following vectors

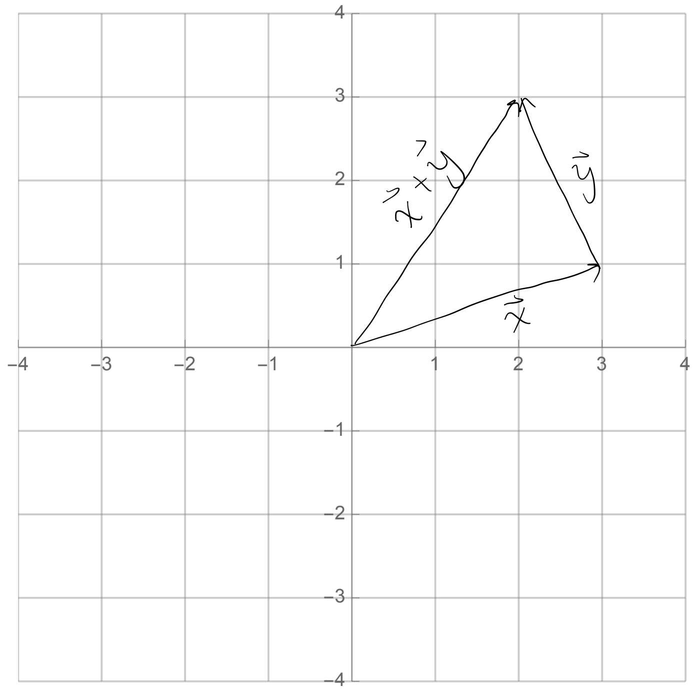
$$\vec{x} = \begin{bmatrix} 3 \\ 1 \end{bmatrix} \qquad \vec{y} = \begin{bmatrix} -1 \\ 2 \end{bmatrix}.$$

a. (5 points) Compute $\vec{x} + \vec{y}$ and illustrate the result graphically.

$$\vec{x} + \vec{y} = \begin{bmatrix} 3 \\ 1 \end{bmatrix} + \begin{bmatrix} -1 \\ 2 \end{bmatrix}$$

$$= \begin{bmatrix} 3-1 \\ 1+2 \end{bmatrix}$$

$$\boxed{\vec{x} + \vec{y} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}}$$



b. (10 points) Find the angle between the two vectors.

$$\vec{x} \cdot \vec{y} = (3)(-1) + (1)(2) = -3+2 = -1$$

$$|\vec{x}| = \sqrt{3^2 + 1^2} = \sqrt{10}$$

$$|\vec{y}| = \sqrt{(-1)^2 + 2^2} = \sqrt{5}$$

$$\boxed{\theta = \cos^{-1}\left(\frac{-1}{\sqrt{50}}\right)}$$

2. (15 points) Find all equilibria and determine the stability of the Levins model given by

$$\frac{dp}{dt} = 4p(1-p) - p.$$

$$\begin{aligned} g(p) &= 4p(1-p)-p \\ &= 4p(1-p)-p \\ &= 4p-4p^2-p \\ &= 3p-4p^2 \end{aligned}$$

$$\underbrace{p}\,(\underbrace{3-4p}) = 0$$

$p=0$ $\quad$ $3-4p=0$

$4p=3$

$p=\frac{3}{4}$

$g'(p) = 3-8p$

$g'(0) = 3 > 0 \Rightarrow \boxed{p=0 \text{ is unstable}}$

$g'(\frac{3}{4}) = 3-8(\frac{3}{4}) = 3-6 = -3 < 0 \Rightarrow \boxed{p=\frac{3}{4} \text{ is stable}}$

3. (15 points) Find a 3rd-degree Taylor polynomial of $f(x) = \sin x$ centered around $x = 0$.

$f(x) = \sin x \qquad f(0) = \sin 0 = 0$

$f'(x) = \cos x \qquad f'(0) = \cos 0 = 1$

$f''(x) = -\sin x \qquad f''(0) = -\sin 0 = 0$

$f'''(x) = -\cos x \qquad f'''(0) = -\cos 0 = -1$

$P_3(x) = 0 + 1(x-0)^1 + \frac{0}{2!}(x-0)^2 - \frac{1}{3!}(x-0)^3$

$\boxed{P_3(x) = x - \frac{1}{3}x^3}$

4. (15 points) Laboratory mice are fed with a mixture of two foods that contain two essential nutrients. Food 1 contains 2 units of nutrient A and 3 units of nutrient B per ounce; food 2 contains 4 units of nutrient A and 2 units of nutrient B per ounce. Each mouse requires 16 units of nutrient A and 12 units of nutrient B per day. How much of each food should one mouse be fed in one day?

x - ounces of food 1

y - ounces of food 2

nutrient A: $2x + 4y = 16$

nutrient B: $3x + 2y = 12$

$$\left[\begin{array}{cc|c} 2 & 4 & 16 \\ 3 & 2 & 12 \end{array}\right] \xrightarrow{3R_1 - 2R_2 \to R_2} \left[\begin{array}{cc|c} 2 & 4 & 16 \\ 0 & 8 & 24 \end{array}\right]$$

$8y = 24$

$y = 3$

$2x = 16 - 4(3)$

$= 16 - 12$

$= 4$

$x = 2$

$$\boxed{\begin{array}{l} \text{2 oz. of food 1} \\ \text{3 oz. of food 2} \end{array}}$$

5. (20 points) Find the eigenvalues and eigenvectors of the matrix

$$A = \begin{bmatrix} 4 & 1 \\ 3 & 2 \end{bmatrix}$$

$\tau = 4+2 = 6$

$\Delta = (4)(2)-(1)(3) = 8-3=5$

$\lambda^2 - 6\lambda + 5 = 0$

$(\lambda-1)(\lambda-5) = 0$

$\underline{\lambda = 1:}$

$$\begin{bmatrix} 4-1 & 1 \\ 3 & 2-1 \end{bmatrix}\begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 1 \\ 3 & 1 \end{bmatrix}\begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$3u_1 + u_2 = 0$

$3u_1 = -u_2$

$$\vec{u} = \begin{bmatrix} -1 \\ 3 \end{bmatrix}$$

$\underline{\lambda = 5:}$

$$\begin{bmatrix} 4-5 & 1 \\ 3 & 2-5 \end{bmatrix}\begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} -1 & 1 \\ 3 & -3 \end{bmatrix}\begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$-v_1 + v_2 = 0$

$v_1 = v_2$

$$\vec{v} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\boxed{\begin{array}{ll} \lambda = 1 & \vec{u} = \begin{bmatrix} -1 \\ 3 \end{bmatrix} \\ \lambda = 5 & \vec{v} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \end{array}}$$

6. (70 points) An experimental flask contains a saltwater solution. There are two pipes connected to the flask. One pours a saltwater solution into the flask and the other pumps the well-mixed solution out of the flask.

a. (10 points) The rate of solution being poured into the flask in L/min. is given to be $9t^2 - 10t + 11$ and the rate of solution being pumped out of the flask in L/min. is given to be $6t^2 - 8t + 10$ where t is in minutes. Given that the flask initially holds 3 L, show that the volume after t minutes is given by the equation

$$V(t) = t^3 - t^2 + t + 3.$$

$$\frac{dV}{dt} = (9t^2 - 10t + 11) - (6t^2 - 8t + 10)$$

$$\frac{dV}{dt} = 3t^2 + 2t + 1$$

$$\int dV = \int (3t^2 - 2t + 1)\,dt$$

$$V = t^3 - t^2 + t + C$$

$$V(0) = 3 = 0^3 - 0^2 + 0 + C$$

$$C = 3$$

$$\boxed{V(t) = t^3 - t^2 + t + 3}$$

b. (10 points) The concentration of the incoming saltwater solution in g/L is given by the formula $C(t) = \dfrac{e^t}{(t+1)^4(t^2-2t+3)}$. Using the fact that $t^3 - t^2 + t + 3 = (t+1)(t^2-2t+3)$ and the information from Part 6a, show that the system can be modeled by the differential equation

$$\frac{dS}{dt} + \frac{6t^2-8t+10}{(t+1)(t^2-2t+3)}S = \frac{e^t(9t^2-10t+11)}{(t+1)^4(t^2-2t+3)}.$$

$$\frac{dS}{dt} = \begin{bmatrix}\text{rate}\\ \text{in}\end{bmatrix} - \begin{bmatrix}\text{rate}\\ \text{out}\end{bmatrix}$$

rate in: $\left[\dfrac{e^t}{(t+1)^4(t^2-2t+3)}\right](9t^2-10t+11)$

rate out: $\left[\dfrac{S}{(t+1)(t^2-2t+3)}\right](6t^2-8t+10)$

$$\frac{dS}{dt} = \frac{e^t(9t^2-10t+11)}{(t+1)^4(t^2-2t+3)} - \frac{6t^2-8t+10}{(t+1)(t^2-2t+3)}S$$

$$\boxed{\frac{dS}{dt} + \frac{6t^2-8t+10}{(t+1)(t^2-2t+3)}S = \frac{e^t(9t^2-10t+11)}{(t+1)^4(t^2-2t+3)}}$$

c. (20 points) Use partial fraction decomposition and solve a system of equations to show that

$$\frac{6t^2 - 8t + 10}{(t+1)(t^2 - 2t + 3)} = \frac{4}{t+1} + \frac{2t - 2}{t^2 - 2t + 3}.$$

$$\frac{6t^2-8t+10}{(t+1)(t^2-2t+3)} = \frac{A}{t+1} + \frac{Bt+C}{t^2-2t+3}$$

$$6t^2-8t+10 = At^2-2At+3A+(Bt+C)(t+1)$$

$$\underline{6t^2}\,\underline{-8t}+\underline{10} = \underline{At^2}-\underline{2At}+\underline{3A}+\underline{Bt^2}+\underline{Bt}+\underline{Ct}+\underline{C}$$

$$\begin{aligned} A + B &= 6 \\ -2A + B + C &= -8 \\ 3A + C &= 10 \end{aligned} \qquad \left[\begin{array}{ccc|c} 1 & 1 & 0 & 6 \\ -2 & 1 & 1 & -8 \\ 3 & 0 & 1 & 10 \end{array}\right]$$

$$\xrightarrow[3R_1 - R_3 \to R_3]{2R_1 + R_2 \to R_2} \left[\begin{array}{ccc|c} 1 & 1 & 0 & 6 \\ 0 & 3 & 1 & 4 \\ 0 & 3 & -1 & 8 \end{array}\right] \xrightarrow{R_2 - R_3 \to R_3} \left[\begin{array}{ccc|c} 1 & 1 & 0 & 6 \\ 0 & 3 & 1 & 4 \\ 0 & 0 & 2 & -4 \end{array}\right]$$

$$2C = -4 \qquad 3B - 2 = 4 \qquad A + 2 = 6$$

$$C = -2 \qquad 3B = 6 \qquad A = 4$$

$$B = 2$$

$$\boxed{\frac{6t^2-8t+10}{(t+1)(t^2-2t+3)} = \frac{4}{t+1} + \frac{2t-2}{t^2-2t+3}}$$

d. (30 points) If there are initially 3 g of salt in the flask, use the differential equation in Part 6b and the answer from Part 6c to show that the amount of salt in the flask is given by the equation

$$S(t) = \frac{(9t^2 - 28t + 39)e^t - 30}{(t+1)^4(t^2 - 2t + 3)}.$$

$$\int p(t)\,dt = \int \frac{4}{t+1}\,dt + \int \frac{2t-2}{t^2-2t+3}\,dt$$

$$= 4\ln|t+1| + \ln|t^2-2t+3|$$

$$I(t) = e^{\int p(t)\,dt} = e^{\ln|(t+1)^4(t^2-2t+3)|} = (t+1)^4(t^2-2t+3)$$

$$S(t) = \frac{1}{(t+1)^4(t^2-2t+3)} \int \cancel{(t+1)^4}\cancel{(t^2-2t+3)} \frac{e^t(9t^2-10t+11)}{\cancel{(t+1)^4}\cancel{(t^2-2t+3)}}\,dt$$

$$= \frac{1}{(t+1)^4(t^2-2t+3)} \int (9t^2-10t+11)e^t\,dt$$

	u	dv
$+$	$9t^2-10t+11$	e^t
$-$	$18t-10$	e^t
$+$	18	e^t
		e^t

$$= \frac{1}{(t+1)^4(t^2-2t+3)}\left[\left(9t^2-10t+11-(18t-10)+18\right)e^t + C\right]$$

$$S(t) = \frac{(9t^2-28t+39)e^t + C}{(t+1)^4(t^2-2t+3)}$$

$$S(0) = \frac{(9\cdot 0^2 - 28\cdot 0 + 39)e^0 + C}{(0+1)^4(0^2 - 2\cdot 0 + 3)}$$

$$= \frac{39 + C}{3} = 3$$

$$39 + C = 9$$

$$C = -30$$

$$\boxed{S(t) = \frac{(9t^2-28t+39)e^t - 30}{(t+1)^4(t^2-2t+3)}}$$